Lecture 02

Question 1 (5 Marks)

Differentiate between indicating, recording, and integrating instruments, providing a specific example for each category.

Answer (Write in Exam):

Measurement instruments are classified into three categories:

1. Indicating Instruments

These instruments display the instantaneous value of the measured quantity at the time of measurement.

Example: Ammeter, Voltmeter.

2. Recording Instruments

These instruments continuously record the measured quantity over a period of time.

Example: Recording Thermometer, Strip Chart Recorder.

3. Integrating Instruments

These instruments measure and record the total quantity over a specified period.

Example: Energy Meter (kWh Meter).

Conclusion:

Indicating instruments show the present value, recording instruments store values over time, and integrating instruments measure the accumulated value.

বাংলায় বুঝার জন্য:

- **Indicating** = ঐ মুহূর্তের মান দেখায়।
- **Recording** = সময়ের সাথে সাথে মান রেকর্ড করে।
- **Integrating** = মোট পরিমাণ হিসাব করে।

Question 2 (5 Marks)

Explain the two major technical disadvantages or risks associated with a simple multi-range ammeter switching arrangement.

Answer (Write in Exam):

The two major disadvantages of a simple multi-range ammeter switching arrangement are:

1. Risk of Meter Damage

During switching, the meter movement may momentarily remain without a shunt resistor, causing excessive current to flow through the meter and potentially damaging it.

2. Variation in Accuracy

Different shunt resistors and switch contact resistances may introduce errors, reducing the measurement accuracy.

Conclusion:

Simple multi-range ammeters are susceptible to damage during switching and may suffer from reduced accuracy.

বাংলায় বুঝার জন্য**:**

- Switch পরিবর্তনের সময় meter পুড়ে যেতে পারে।
- বিভিন্ন shunt ও switch resistance-এর কারণে error হয়।

Question 3 (1 Mark)

Recall the significance of shunt resistance in an ammeter.

Answer (Write in Exam):

The shunt resistance in an ammeter allows most of the current to bypass the meter movement, thereby extending the current measuring range and protecting the instrument.

বাংলায় বুঝার জন্য**:**
Shunt resistor মূল current-এর বেশিরভাগ অংশ নিজের মাধ্যমে প্রবাহিত করে meter-কে রক্ষা করে।

Question 4 (5 Marks)

What is an Ayrton shunt (or universal shunt)? Explain how it successfully eliminates the structural limitations of a simple multi-range ammeter.

Answer (Write in Exam):

An **Ayrton shunt**, also known as a **universal shunt**, is a special arrangement of resistors used in multi-range ammeters where the meter movement always remains connected to a shunt resistor.

Unlike a simple multi-range ammeter, the Ayrton shunt prevents the meter from being disconnected from the shunt during range switching.

Advantages:

- Protects the meter movement from excessive current.
- Eliminates the possibility of open-circuit damage during switching.
- Provides better measurement reliability and safety.

Conclusion:

The Ayrton shunt overcomes the structural limitations of simple multi-range ammeters by ensuring continuous shunt protection.

বাংলায় বুঝার জন্য:
Ayrton shunt-এর সবচেয়ে বড় সুবিধা হলো switch ঘোরানোর সময়ও meter কখনো shunt ছাড়া থাকে না, তাই meter নষ্ট হওয়ার ঝুঁকি থাকে না।

Question 5 (5 Marks)

Explain what "meter loading" means in the context of an ammeter and how it alters the circuit being tested.

Answer (Write in Exam):

Meter loading in an ammeter refers to the effect caused by the internal resistance of the ammeter when it is inserted into a circuit.

Since the ammeter has a small internal resistance, adding it to the circuit slightly increases the total circuit resistance. This causes a reduction in circuit current, resulting in a measurement that differs from the actual current.

Conclusion:

Meter loading alters the circuit conditions and introduces measurement errors.

বাংলায় বুঝার জন্য:
Ammeter circuit-এর মধ্যে ঢুকলে circuit-এর resistance একটু বেড়ে যায়, ফলে আসল current কিছুটা কমে যায়।

Question 6 (5 Marks)

What is the structural and mathematical purpose of connecting multiplier resistors in series with a basic PMMC movement?

Answer (Write in Exam):

Multiplier resistors are connected in series with a PMMC movement to extend its voltage measuring range.

Structural Purpose:

The series resistor limits the current flowing through the PMMC movement and protects it from excessive current.

Mathematical Purpose:

The multiplier resistor increases the total resistance according to Ohm's law:

$$R_M = \frac{V}{I} - R_g$$

where,

- R_M = Multiplier resistance,
- V = Desired voltage range,
- I = Full-scale deflection current,
- R_g = Internal resistance of the meter.

Conclusion:

Multiplier resistors protect the PMMC movement and extend its voltage measuring capability.

বাংলায় বুঝার জন্য**:**
Multiplier resistor series-এ বসিয়ে current কমানো হয়, যাতে ছোট PMMC meter দিয়ে বড় voltage মাপা যায়।

Question 7 (5 Marks)

Describe the internal resistance property of an ideal voltmeter and explain why perfect voltmeters only exist in textbooks.

Answer (Write in Exam):

An ideal voltmeter has infinite internal resistance.

Because of this infinite resistance, it draws zero current from the circuit and does not affect the voltage being measured.

However, in practical situations, every voltmeter has a finite internal resistance due to physical and material limitations. Therefore, practical voltmeters always draw a small amount of current and introduce some measurement error.

Conclusion:

Perfect voltmeters exist only theoretically because achieving infinite resistance is physically impossible.

বাংলায় বুঝার জন্য:

Ideal voltmeter-এর resistance অসীম হওয়া উচিত, যাতে circuit থেকে কোনো current না নেয়। কিন্তু বাস্তবে কোনো যন্ত্রের resistance অসীম হতে পারে না, তাই perfect voltmeter শুধু বইতেই আছে।